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Assignment 2

Questions

1. Why does BFS ignore the distances stored in the graph even though they are available to it?

BFS ignore the distance stored in the graph because it finds the fewest steps or edges to the goals. It searches all nodes level by level until it reaches the goals. And then it selects shallowest solution based on the fewer edges to the goals without looking at the distances or weight.

2. Which two lines change when bfs is turned into dfs, and which of them makes DFS order dependent?

The two lines that change are the frontier data structure initialization (Queue becomes a Stack) and the removal operation (popping from the front becomes popping from the top/end). The LIFO removal operation (popping from the top) makes DFS order-dependent.

3. UCS tests for the goal when a state is popped from the heap. What could go wrong if it tested when the state was first generated?

If UCS tests for the goal when a state is generated rather than popped, it may return a sub-optimal path. A node's shortest path is only guaranteed when it is popped from the priority queue, not when it is first discovered.

4. Lab1reports *expanded* and *runtime_ms*. Which one should you use to compare BFS and UCS with a classmate's run on a different laptop, and why?

You should use expanded (nodes expanded). runtime_ms is hardware-dependent and will vary between laptops, whereas the number of nodes expanded is a deterministic mathematical property of the algorithm itself.

5. Which quantity does GBFS ignore that A⋆ uses, and which failure in Lab2 does that explain?

GBFS ignores the accumulated path cost from the start node, $g(n)$. It only relies on the heuristic, $h(n)$, which explains why GBFS fails to guarantee the optimal/shortest path in Lab2, as it greedily prioritizes estimated closeness to the goal over the actual cost incurred.

6. Frontier states *X* and *Y* have *(g, h) = (8,2)* and *(3,6)*. Which does UCS expand first? GBFS? A⋆?

- **UCS** uses $g(n)$. It expands **Y** first (3 < 8).
- **GBFS** uses $h(n)$. It expands **X** first (2 < 6).
- **A*** uses $f(n) = g(n) + h(n)$. $f(X) = 10$, $f(Y) = 9$. It expands **Y** first (9 < 10).

7. Why must *h* be expressed in the same units as the edge costs?

A* calculates the total estimated cost as $f(n) = g(n) + h(n)$. To mathematically add these two values together, they must be in the exact same units (both must be distance or both must be time).

8. Lab2 computes $h\star$ by running Dijkstra on the reversed graph from the goal. Why reversed?

Dijkstra's algorithm finds the shortest path *from* a single source *to* all other nodes. By reversing the graph's edges and running it from the goal node, it calculates the exact true cost from all nodes *to* the goal, yielding the perfect heuristic h^*

9. Weighted A⋆ with $w = 1.5$ changed the cost for KUL→KEF but not for SIN→EZE. What property of the SIN→EZE route explains this?

If inflating the heuristic ($w = 1.5$) did not change the cost or route for SIN→EZE, it indicates that the optimal path is highly constrained or strictly dominant. There were no competitive alternative routes for the inflated heuristic to misguide the algorithm into exploring